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وزارة التربية والتعليم والتعليم الفني
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Student guide

Physics

Year 2

(STEM)

Prepared by
Saeed Mohamed Ali
Revised by
Dr. Aziza Ragab Khalifa

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Unit one

Force

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Dear student, By the end of this lessons you should have the following skills and knowledges

Understand the concept of force

Derive the international unit of force

State some types of force.

Apply the universal Gravitation law.

Identify how to find the error in measurement.

Identify the reasons of centripetal force, the friction force and the force of viscosity.

Differentiate between types of force

Derive the international unit of viscosity coefficient

Deduce the factors affecting the different types of force

Lesson 1

Meaning of force and its measuring units

Introduction to force

When you kick the ball, you exert a force. When you sit in the chair, you exert a force. When you push or pull the door, you exert a force also. Push or pull of an object is considered a force. Push and pull come from the objects interacting with one another. **So, forces describe the interactions between an object and its environment.**

When there is an **interaction** between two objects, there is a force upon each of the objects. When **the interaction vanishes**, there is no force between the two objects. **Forces only exist as a result of an interaction.**

The concept of force F

- (An external effect that affects the object, it changes or attempts to change its state or direction).
- (An action which produces or tends to produce a change in a body's state of rest or motion).

Effects of the Force

The Force has different effects such as

- a) **The Force** can make a body that is at rest to move.
- b) **The Force** can stop a moving body or slow it down.
- c) **The Force** can accelerate the speed of a moving body.
- d) **The Force** can change the direction of a moving body along with its shape and size.

Types of forces:

Force can be classified into two broad categories

A) Contact forces

These types of forces are produced when two objects interact with each other; they have a physical contact with each other

Examples of contact forces

- a) **Frictional force**
- b) **Tension force**
- c) **Normal Force**
- d) **Air Resistance Force**
- e) **Applied Force**
- f) **Spring Force**

B) Action at a distance force (Non-Contact forces)

These types of forces happen when two interactive objects are not in physical contact with each other.

Examples of Action at a distance force (Non-Contact forces)

- a) **Gravitational force**
- b) **Electrical force**
- c) **Magnetic forces**

Measuring units of force

The relationship between force and acceleration will be discussed by studying Newton's laws of motion that are considered as basic laws in physics.

We have learnt that when no resultant force affects the object, it does not move at acceleration. Consequently, when a resultant force acts on the object ($\Sigma F \neq 0$), it moves at acceleration. Accordingly, its velocity changes and it acquires acceleration ($a \neq 0$)

The force has a magnitude and a direction. The direction towards which the force is applied

$$\text{Force} = \text{Mass} \times \text{Acceleration}$$

$$F = m a$$

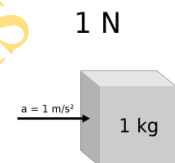
Force is a vector quantity (magnitude & direction) that is measured using the standard metric unit (SI unit) known as the **Newton**

Newton is equivalent to $\text{kg} \cdot \text{m/s}^2$

Definition of Newton

$$\text{Force} = \text{Mass} \times \text{Acceleration}$$

$$\text{Newton} = \text{kg} \times \text{m/s}^2$$



(It is the force acting on an object of mass 1 kg to give acceleration 1 m/s²)

Solved problem:

1- A truck with a mass of 2000 kg is decelerated at a rate of 7 m/s². calculate the force acting on the truck

$$F = m a$$

$$F = 2000 \times 7 = 14000 \text{ N}$$

Lesson 2

Gravitational force

What Is Gravity?

Gravity is the force by which a planet or other body draws objects toward its center. The force of gravity keeps all of the planets in orbit around the sun.

What else does gravity do?

Why do you land on the ground when you jump up instead of floating off into space?

Why do things fall down when you throw them or drop them?

The answer is **gravity that is an invisible force that pulls objects toward each other.**

Anything that has mass also has gravity. Objects with more mass have more gravity.

Gravity also gets weaker with distance.

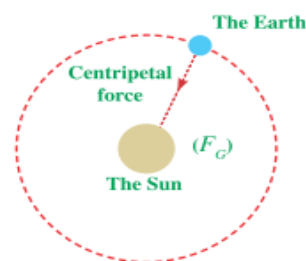
Gravity is very important to us. We could not live on Earth without it. The sun's gravity keeps Earth in orbit around it .

Gravitational force (Force of gravity) (Fg):

- a) A gravitational force exists between **the Earth or moon or other massively huge objects attract another object towards them.**

This force is normal to the direction of motion of the Earth

- b) **All objects on the Earth experience the gravitational force, which is directed downwards towards the center of the earth**



Definition of Gravitational force

The gravitational force, can be defined as the force of attraction that arises between two bodies having definite masses which is directly proportional to the product of the masses of the two bodies and inversely proportional to the distance between them squared

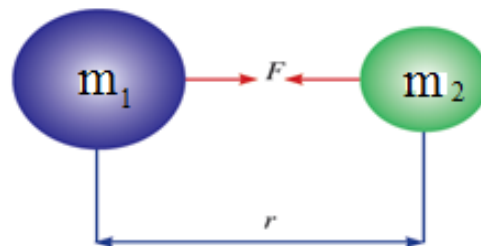
The mathematical formula of universal Gravitation law

$$F = \frac{G m_1 m_2}{r^2} = mg$$

G: the universal constant ($G = 6.67 \times 10^{-11}$ N.m²/kg²)

m₁, m₂: the masses of the two bodies

r: the distance between the centers of the two bodies



The force acting on first body = the force acting on second body

i.e. the attraction force is a mutual force between the bodies, each one attracts the other by the same force

Solved problem

Two small balls of mass (7.3 kg) are placed at a distance between their centers equal to (0.5 m) calculate the force of attraction between them

The solution

$$F = \frac{G m_1 m_2}{r^2} = \frac{6.67 \times 10^{-11} \times 7.3 \times 7.3}{0.5^2} = 1.4 \times 10^{-8} \text{ N}$$

Exercise

If the distance between the centers of two identical spheres is 1m, and the force of attraction between them is 1N, then calculate the masses of the two spheres.

Application of Gravitational force

The law of gravitation can be used to explain the followings:

- a) The motion of planets around the sun
- b) The motion of the moon (satellite) around the earth
- c) The motion of a freely falling body

Applied Technology Schools

Lesson 3

Centripetal force

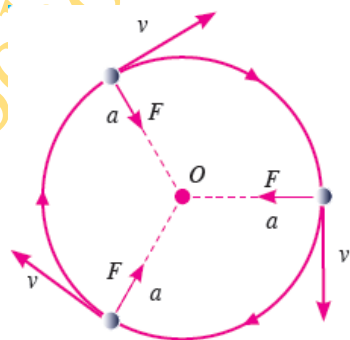
Introduction

Motion in a circle (motion that repeats itself at equal intervals of time such as the motion of the Earth around its axis and also around the sun) is one of the most important common motions in nature. Examples include the motion of some games in amusement parks, the motion of Earth around the Sun, the moon around Earth, etc. Accordingly, we are concerned in this unit with the study of motion in a circle, describing its mechanism, illustrating a number of its life examples, deducing the mathematical relations used to express it, and demonstrating most important life and technological applications.

Through your study to Newton's second law, you learned that when a force acts on a body moving at a certain velocity, the body acquires an acceleration, i.e.

A change in its velocity takes place. This change depends on the direction of the acting force relative to the direction of motion as follows:

- 1- The direction of force is in the same direction of motion
The magnitude of velocity increases & its direction does not change
- 2- The direction of force is in the opposite direction of motion. The magnitude of velocity decreases & its direction does not change
- 3- The direction of force is perpendicular to direction of motion. The direction of velocity changes & its magnitude does not change



When an object moves in a circular path with a constant velocity (the object moves in a uniform circular motion), the acceleration of the object is always directed towards the center of the circular path and perpendicular to the direction of the instantaneous velocity and the **centripetal** acceleration (a_c) is determined from the relation

$$a_c = \frac{v^2}{r}$$

v : velocity of the moving object in the circular path (m/s)

r : radius of the circular path (m)

Uniform circular motion: it is the motion of an object in a circular path at a constant speed but changeable direction. The force acting on this object towards the center is called the centripetal force.

Centripetal acceleration (a): It is the acceleration acquired by the object in a circular motion due to a change in the direction of its velocity.

Centripetal force: it is the force acting continuously in a direction normal to the motion of a body, **accordingly**, its path changes from being straight into circular.

Calculating of the centripetal force:

$$F_c = m a_c = \frac{m v^2}{r}$$

The factors affecting the centripetal force:

It is necessary to calculate the centripetal force when designing a curve in roads or railways in order to help cars and trains move along this curved path without skidding.

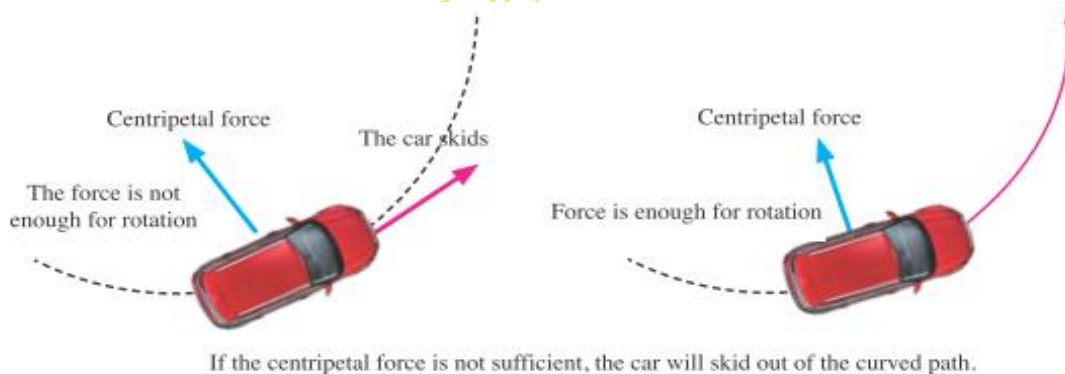
Through studying the relation, we can conclude that the centripetal force depends on each of the following factors:

- 1 - **Mass of the object (m):** where the centripetal force is directly proportional to mass at constant (v, r). The force required to move a bicycle in a curved path is less than that required to move a truck in the same path. This explains why trucks are not allowed to move along some dangerous curves.
- 2- **Tangential velocity (v):** where the centripetal force is directly proportional to the square of velocity at constant (r, m). When the velocity of a car increases, it needs more centripetal force to move within a curved path. So engineers define a certain velocity which should not be exceeded for the motion in curves road and highway.
- 3- **Radius of rotation (r):** where the centripetal force is inversely proportional to the radius of the path at constant (v, m). i.e, when the radius of curve decreases, the car needs more centripetal force to keep moving in a circle. Accordingly the curve becomes more dangerous. To avoid sliding out, it is advisable to move with low speeds in dangerous curves.

What will happen if the centripetal force suddenly becomes zero = zero?

The resultant force acting on the body will be $= 0$, therefore the acceleration acquired by the object $= 0$ and then the object will move with a uniform velocity in a straight line tangent to the circular path according to Newton's first law.

Assume that a car moved on a curved path, and the road has oil stains, friction forces are not enough to rotate the car within the curved path which causes the car to skid. The tires creep on the bypath and the car cannot keep moving in the curved path.



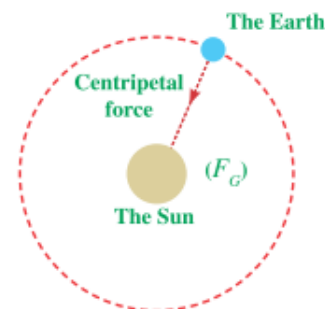
Notes

The centripetal acceleration a_c can be expressed in terms of the periodic time T , where the periodic time is the time required to make one complete cycle

$$v = \frac{d}{t} = \frac{2\pi r}{T}$$

$$a_c = \frac{v^2}{r} = \frac{4\pi^2 r}{T^2}$$

Centripetal force is not considered as a new type of force. Simply, it is the term given to any force acting normally to the path of moving an object and makes it move in a circular path. May the centripetal force be tension force, or gravitational force or Friction force or Reaction force or Lifting force.

**Solved problems**

1) An object of mass 2kg is tied at the end of a string to rotate in a horizontal circular path of radius 1.5m at a uniform linear velocity of 28.26m/s. calculate

- a) the centripetal acceleration b) the tension of the string of object

The solution

$$\therefore a_c = \frac{v_o^2}{r}$$

$$\therefore a_c = \frac{(28.26)(28.26)}{(1.5)} = \frac{(798.6276)}{(1.5)}$$

$$\therefore a_c = 532.42 \text{ m/s}^2$$

$$\therefore F = m a_c = (2)(532.42)$$

$$\therefore F = 1064.84 \text{ N}$$

2) A toy of mass 600g is moving around the circumference of a circle of diameter 4m with a constant linear velocity of 10m/s Calculate

- a) the centripetal acceleration b) the centripetal force
c) Time of one revolution (periodic time)

$$\therefore a_c = \frac{v_o^2}{r}$$

$$\therefore a_c = \frac{(10)(10)}{(2)} = 50 \text{ m/s}^2$$

$$\therefore F_c = m a_c$$

$$\therefore F_c = (0.6)(50) = 30 \text{ N}$$

$$\therefore T = \frac{2\pi r}{v_o} = \frac{(2)(3.14)(2)}{(10)} = 1.256 \text{ s}$$

Lesson 4

Friction force

Introduction

As an object moves across a surface it causes friction. Friction force can be sliding or static. Friction depends upon the nature of the two interacting surfaces. Such as

- 1) A book sliding on the table,
- 2) A ball rolling on the floor.

Concept of friction

Friction is the force that resists motion when the surface of one object comes in contact with the surface of another.

Frictional force is the force generated by two surfaces that contact and slide against each other.

Frictional force is the force that resists the sliding or rolling of one solid object over another

The factors affecting the frictional force:

- a) The force with which the two surfaces are pressed together
- b) The nature of the two surfaces that are sliding across each other
- c) The angle and position of the object
 - 1) If an object is placed flat against an object, then the frictional force will be equal to the weight of the object
 - 2) If an object is pushed against the surface, then the frictional force will be increased and becomes more than the weight of the object

Types of friction force

Friction is the force that opposes motion between any surfaces that are in contact.

The main types of friction

A- Static friction

The static friction occurs between solid surfaces

The force of static friction F_s is a force between two surfaces that prevents those surfaces from sliding or slipping across each other. i.e. Static friction acts on objects when they are resting on a surface

Examples:

- 1) When you are hiking in the woods, there is static friction between your shoes and the trail each time you put down your foot
- 2) the friction that helps the people climb the vertical rock wall



- 3) Your planted foot can grip the ground and push backward, which causes the ground to push forward on your foot.

Without the static friction, your feet would slip out from under you, making it difficult to walk. In fact, that's exactly what happens if you try to walk on ice. That's because ice offers very little friction.

B- Sliding Friction

Sliding friction occurs also between solid surfaces.

Sliding friction is friction that acts on objects when they are sliding over a surface.

Examples

- 1) When you write with a pencil. The pencil "lead" slides easily over the paper but there's just enough friction between the pencil and paper to leave a mark.
- 2) When you slide a piece of furniture over the floor after you start it moving than it is to get it moving in the first place.
- 3) The sliding friction between the brake pads and bike rims each time you use your bike's brakes. This friction slows the rolling wheels so you can stop.

C- Rolling Friction

Rolling friction occurs also between solid surfaces.

Rolling friction is friction that acts on objects when they are rolling over a surface.

Examples

There are many forms of ground transportation use wheels

- 1) Bicycles,
- 2) cars,
- 3) 4-wheelers, roller skates, scooters, and skateboards.
- 4) Ball bearings are another use of rolling friction.

They let parts of a wheel or other machine roll rather than slide over on another. The ball bearings in this wheel reduce friction between the inner and outer cylinders when they turn.



- a) Sliding and Rolling friction are considered from kinetic friction force that **always opposes sliding motion and tries to reduce the speed with which the surfaces slide through each other.**
- b) Sliding friction is weaker than static friction.
- c) Rolling friction is much weaker than sliding friction or static friction

D- Fluid friction

Fluid friction occurs in liquids and gases.

Fluid friction is friction that acts on objects that are moving through a fluid (**a substance that can flow**)

Examples

- 1) **When you** tried to push your open hand through the water in a tub or pool, then you've experienced fluid friction (called water resistance)
- 2) In the opposite **Figure**, the skydiver is falling toward Earth with a parachute (called air resistance against the parachute slows his descent).



The faster or larger a moving object is, the greater is the fluid friction resisting its motion. That's why there is greater air resistance against the parachute than the skydiver's body.

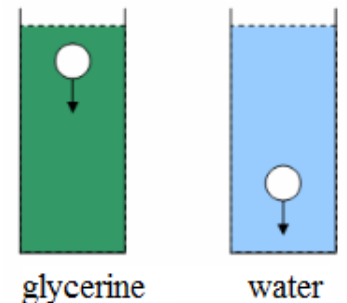
Lesson 5

Viscosity pheromone and lubrication process

Viscosity

We observe viscosity as follows:

- 1) Take two similar glass beakers one containing a certain volume of water and the other an equal volume of glycerine
- 2) Take two similar metallic balls drop one of them in water and record the time taken by the ball to reach the bottom of the cylinder
- 3) Drop the second ball in glycerin and record the time taken to reach the bottom

**Observation:**

The time record in case of water is less than that in case of glycerin.

Conclusion:

The glycerin resistance to the ball motion is greater than that of water, so glycerin more viscous than water.

Viscosity

It is the property of fluid due to existence of friction force between each liquid layer and the adjacent one, which resists the sliding of the liquid layers with respect to each other. This produces a relative change in velocity between any two adjacent layers. Thus, viscosity is the property responsible for resisting the relative motion of liquid layers.

Note that

- 1- The most viscous liquid: it flows slowly and offer high resistance to the movement of objects in it.
- 2- The least viscous liquid: it flows quickly and offer little resistance to the movement of objects in it.

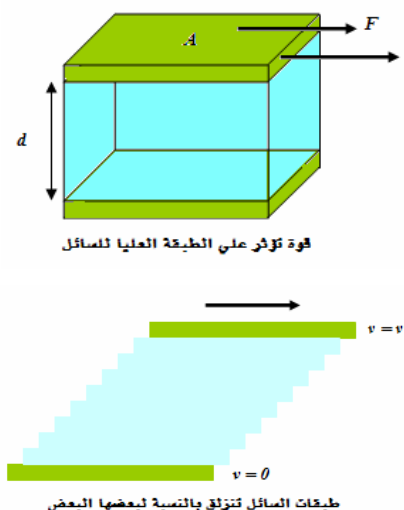
The factors on which the viscosity coefficient of a liquid depends:

Viscosity is a physical property, so it depends on:

- 1- The type of liquid.
- 2- The temperature.

Interpretation of viscosity in light of the theory of laminar flow

- In order to explain the viscosity characteristic, we imagine a drop of liquid confined between two plane plates with a distance (d) between them, one of them is at rest and the other is moving at a speed v.
- The liquid layer in contact with the stationary plate is stationary, while the liquid layer in contact with the moving plate moves with the same velocity, which is (v), and the liquid layers between the two plates move at speeds from zero to v.
- It increases from the static plate to the movable plate, where the velocity of each layer is less than the speed of the layer above it.



The relative change in velocity is due to the following:

- Friction forces exist between the lower plate and the liquid layer in contact with it. This force is due to the adhesive forces between the molecules of the solid surface and the contacting liquid molecules. This leads to zero velocity of the layer in contact with the stationary plate. Similarly, the upper layer moves at the same velocity of the upper plate.
- The existence of another friction (shear) force between each liquid layer and the adjacent one, which resists the sliding of the liquid layers with respect to each other. Thus, viscosity is the property responsible for resisting the relative motion of liquid layers

The factors affecting the viscosity forces and the relation between them and deduction of the viscosity coefficient

Referring to the previous figure, we find that in order for the movable plate to maintain a constant velocity, there must be a tangential force (F_{vs}). This force is directly proportional to both the speed (v) and the area of the movable plate (A) and inversely proportional to the distance between the two plates (d).

$$F_{vs} = \eta_{vs} \frac{AV}{d}$$

η is the coefficient of viscosity (constant of proportionality)

The coefficient of viscosity η :

$$\eta = \frac{F d}{A v}$$

Definition:

It is the tangential force acting on unit area to produce a unit change in velocity between two liquid layers at unit distance apart from each other

The measuring units

1) $N.s / m^2 = N.m^{-2}.s$

2) $kg/m.s = kg.m^{-1}.s^{-1}$

3) Pascal .s

4) $J.s/m^3$

Applications of Viscosity

1-Lubrication:

Metallic parts in machines have to be lubricated from time to time. This process leads to:

- a) **Reduction** of heat generated by friction.
- b) **Protecting** machine parts from corrosion (wear).

Lubrication is carried out using highly viscous liquids, if we use water (low viscosity), it will soon seep away or sputter from the machine parts due its low adhesive forces. Therefore, we must use liquids with high adhesive (high viscosity), so they remain in contact with the moving machine parts.

Solved problem

A flat slab with an area of 0.1 m^2 is placed on a flat surface with a layer of oil separating them with a thickness of 10^{-5} m and coefficient of viscosity 1.5 kg/m.s Find the force required to move the upper plate at a constant speed of 1 m/s .

The solution:

$$\because F_{\text{vs}} = \eta_{\text{vs}} \frac{AV}{d}$$
$$\therefore F_{\text{vs}} = 1.5 \frac{0.1 \times 1}{10^{-5}} = 15000 \text{ N}$$

Unit two

Heat

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Dear student, by the end of this chapter you should have the following skills and knowledges

Differentiate between temperature and heat

Identify the relation between thermal internal energy and temperature

Measure temperature in different scales

Understanding the relation between the length of liquid column with temperature in Liquid thermometer

Differentiate between the heat capacity and the specific heat

Describe how energy changes as a result of energy transfer

Identify the mechanisms of heat transfer

Lesson 1

Temperature and thermal energy

Introduction

The terms (temperature) and (heat) are often used interchangeably in everyday language. In physics, however, these two terms have very different meanings.

We'll define **temperature** in terms of how it's measured and see how temperature changes affect the dimensions of objects. We'll see that **heat** refers to energy transfer caused by temperature differences.

The concept of **temperature** is rooted in qualitative ideas of "hot" and "cold" based on our sense of touch. A body that feels hot usually has a higher temperature than a similar body that feels cold.

Concept of Heat:

(It is thermal energy transferred from hotter system to cooler system that are in contact).

(It is the total energy of the motion of the molecules of a substance).

From the factors affecting Heat, the speed of the particles, the size of the particles and the number of particles

Temperature versus Heat

Temperature depends on the physical state of a material and is a quantitative description of its hotness or coldness.

Heat always refers to energy in transit from one body or system to another because of a temperature difference,

Thermal internal energy:

The Thermal internal energy of a body can be defined as the sum of the kinetic energy and potential energy of its molecules.

Temperature:

(It is a measure of the average kinetic energy of the atoms or molecules in the system).

OR

(It is the measure of the average energy of the motions of the molecules in the substance)

Changing temperature:

Many properties of matter that we can measure depend on changing temperature such as the length of a metal rod, steam pressure in a boiler, the ability of a wire to conduct an electric current, and the color of a very hot glowing object.

Effect of temperature

1) Changing the state of matter

When the temperature increases, the material changes from **solid state to liquid state** for example, change piece of ice into water. Also change water from **liquid state into vapor state**.

2) Most materials expand when their temperatures increase.

When matter gets hot (gain temperature), its molecules go faster and take up more spaces and the solid, liquid or gas expands and vice versa. Rising temperatures make the liquid expand in a liquid-in-tube thermometer; the decks of bridges need special joints and supports to allow for expansion.

The temperature of a body is a measure of its thermal internal energy

Quantity of Heat (Q_{th})

Heat is sometimes called a **process quantity**, because it is defined in the context of a process by which energy can be transferred. We can calculate the heat released or absorbed using the following relation

$$Q_{th} = m C \Delta t$$

m: mass of the object

C: specific heat (it is a physical property of a material different for different materials i.e. does not depend on mass of the object)

Δt : difference (change) in temperature

Solved problem:

Calculate the quantity of heat necessary to raise the temperature of 0.3kg copper from 20°C to 70°C given that the specific heat copper = 390 J/ kg. K.

The solution

$$Q_{th} = m C \Delta t = 0.3 \times 390 \times (70 - 20) = 5850 \text{ J}$$

The relation between thermal internal energy and temperature

When a body gains quantity of heat energy, there is some changes inside the body take place such as the amplitude of molecules vibration increases, the speed of the molecules increases **and** the kinetic energy of the molecules increases. **Therefore** the internal energy of the molecules increases and vice versa.

- 1) **Q and Δt** can be either positive or negative. When they are positive, heat enters the body (gain energy) and its temperature increases; when they are negative, heat leaves the body (lose energy) and its temperature decreases.
- 2) If the temperature of a body is constant but different from that of surrounding medium. Therefore ,the rate of the heat gained by the material equals the rate of the heat lost by the material

Lesson 2

Measuring temperature

Introduction

Need to know if you should put a coat on before you go out? Want to check if you can put the cookies in the oven? Temperature scales provide a way of quantifying and measuring how hot or cold a material is. There are three major temperature scales that are used around the world – **Fahrenheit and Celsius** are frequently used in everyday, while **Kelvin** is more commonly used in the science.

Measurement of temperature:

Temperature is a measure of the amount of heat energy possessed by an object. Because temperature is a relative measurement, scales based on reference points must be used to accurately measure temperature.

Thermometers

To use temperature as a measure of hotness or coldness, we need to construct a temperature scale. To do this, we can use any measurable property of a system that varies with its “hotness” or “coldness. The opposite Figure shows a familiar system that is used to measure temperature.

Thermometers are devices used to measure the temperature of a system. All thermometers are based on the principle that some physical property of a system changes as the system's temperature changes.

Thermal equilibrium

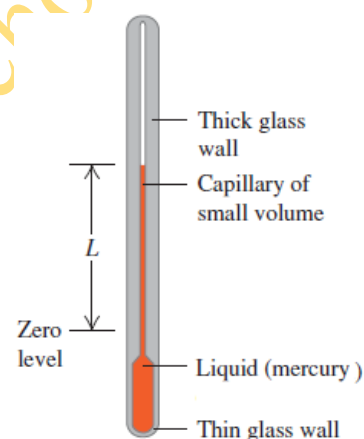
(The state in which two bodies in physical contact with each other have identical temperature)

Thermal equilibrium is the basis for measuring temperature with thermometers. By placing a thermometer in contact with an object and waiting until the column of liquid in the thermometer stops rising or falling, you can find the temperature of the object. This is because the thermometer is in thermal equilibrium.

Liquid thermometer

It is the common thermometer in everyday that consists of a liquid (mercury or alcohol) inside a glass capillary tube. It is based on the principle that is **the regular change of the length of liquid (volume of a liquid) column with temperature**

i.e. Any temperature change in the range of the thermometer can be defined as being proportional to the change in length of the liquid column.



Calculating the temperature that corresponds the length of liquid column in the thermometer

$$t = 100 \times \frac{L_t - L_0}{L_{100} - L_0} = 100 \times \frac{V_t - V_0}{V_{100} - V_0}$$

L_t (V_t): the length (volume) of liquid column at temperature (t)

L_0 (V_0): the length (volume) of liquid column at 0 °C

L_{100} (V_{100}): the length (volume) of liquid column at 100 °C

It is useful to distinguish between an actual temperature (t) and a difference or change in temperature (Δt). For Example, an actual temperature (t) of is stated as 20 °C, and a difference or change in temperature (Δt) of is 20 °C. A beaker of water heated from 30 °C to 50 °C undergoes a temperature change of 20 °C. Actual temperatures are $t_1 = 30$ °C, $t_2 = 50$ °C. But difference or change in temperature (Δt) = 50 - 30 = 20 °C

Solved problem

If the length of mercury in a thermometer = 15 cm at the melting point of ice and the length of mercury in a thermometer = 40 cm at the boiling point of water. Find the temperature at which the length of mercury in a thermometer = 25 cm.

The solution

$$t = 100 \times \frac{L_t - L_0}{L_{100} - L_0} = 100 \times \frac{25 - 15}{40 - 15} = 40 \text{ °C}$$

Temperature scales

There are three main scales commonly used in the world today to measure temperature: the **Fahrenheit** (°F) scale, the **Celsius** (°C) scale, and the **Kelvin** (K) scale. Each of these scales uses a different set of divisions based on different reference points.

A temperature measurement in any of the three scales can be easily converted to another scale using the simple following formulas.

$$T_K = T_c + 273$$

$$T_F = \left(\frac{9}{5} \times T_c \right) + 32$$

Temperature scales	T_C	T_F	T_K
The freezing temperature of pure water (ice point)	0 °C	32 °F	273 K
The boiling temperature of pure water (steam point)	100 °C	212 °F	373 K

Exercises

- On a day when the temperature reaches 50°F, what is the temperature in degrees Celsius and in kelvins?
- Rank (arrange) the following temperatures from highest to lowest:
(a) 0 °C (b) 0 °F; (c) 260 K; (d) 77K ;(e) -180 °C.

3) Choose the correct answer

If the temperature of a person increases by 2 on the Celsius scale, the temperature increases on Kelvin scale by..... K

- a) 2 b) 275 c) 271 d) 371 e) 375

Differences between heat and temperature:

We can remember the importance differences between Heat and Temperature in the following table

Point of comparison	Heat	Temperature
Definition	The heat of an object is the total energy of all the molecular motion inside that object.	Temperature is the measure of the thermal energy or average heat of the molecules in a substance.
Measuring tool	Amount of heat transferred between two bodies can be measured by Calorimeter.	Temperature of a body can be measured by Thermometer.
SI Unit	Joule	Kelvin
Units	Joules, Calories	Fahrenheit, Celsius, Kelvin
Symbol	Q	T
Ability to do Work	It has the ability to do work.	It can be used to measure the degree of heat.

Lesson 3

Heat capacity and specific heat

Introduction

If a swimming pool and wading pool, both full of water at the same temperature, were subjected to the same input of heat energy, the wading pool would certainly rise in temperature more quickly than the swimming pool.

The difference between the heat capacity of a body and the specific heat

Why the wading pool would certainly rise in temperature more quickly than the swimming pool?

To answer the question, we shall understand the difference between the heat capacity of a body and the specific heat

The heat capacity of a body (q_{th})

We can define the heat capacity as follows

(It is the heat quantity required to the whole body to raise its temperature by one Kelvin)

$$q_{th} = \frac{Q_{th}}{\Delta t}$$

q_{th} : the heat capacity of a body measured in (J/ K)

Q_{th} : the quantity of heat measured in **Joules (J)**

Δt : the change of temperature (K)

The heat capacity of a body is not a physical property of the material because the heat capacity of a body depends on its mass.

The specific heat of a material

We can define the heat capacity as follows

(It is the quantity of heat required to raise the temperature of a unit of mass (1kg) of a substance by one Kelvin)

$$C_{th} = \frac{Q_{th}}{m \Delta t} = \frac{q_{th}}{m}$$

C_{th} is the specific heat of a material measured in (J/ kg. K)

m is the mass of a material measured in kilogram (kg)

The specific heat of a body is a physical property of the material because the **specific heat** of a body does not depend on its mass but varies from one material to another. Different substances respond to heat in different ways. If a metal chair sits in the bright sun on a hot day, it may become quite hot to the touch. An equal mass of water under the same sun exposure will not become nearly as hot. This means that water has a high specific heat

The opposite table lists the specific heats capacity of some common substances:

The wading pool would certainly rise in temperature more quickly than the swimming pool because the heat capacity of an object depends both on its mass and its chemical composition, since the swimming pool **has larger mass than** the wading pool despite **both have the same water at the same temperature**, therefore the swimming pool of water has a larger heat capacity than the wading pool.

Substance	Specific Heat (J/g °C)
Water (l)	4.18
Water (s)	2.06
Water (g)	1.87
Ammonia (g)	2.09
Ethanol (l)	2.44
Aluminum (s)	0.897
Carbon, graphite (s)	0.709
Copper (s)	0.385
Gold (s)	0.129
Iron (s)	0.449
Lead (s)	0.129
Mercury (l)	0.140
Silver (s)	0.233

Lesson 4

Heat transfer

Introduction

- 1- When you put a cold spoon into a cup of hot coffee, the spoon warms up and the coffee cools down as they approach thermal equilibrium. The interaction that causes these temperature changes is fundamentally a transfer of energy from one substance to another. Energy transfer that takes place because of a temperature difference is called **heat transfer**, and energy transferred in this way is called **heat**.
- 2- It is well known that any material in the form of a **solid, liquid or gas** is composed of molecules.

Heat transfer

Any matter which is made up of atoms and molecules has the ability to transfer heat. The atoms are in different types of motion at any time. The motion of molecules and atoms is responsible for heat or thermal energy and every matter has this thermal energy. The more the motion of molecules, more will be the heat energy. However, talking about heat transfer, it is nothing but the process of transfer of heat from a high-temperature body to a low temperature one.

- 1) external heat is the thermal energy that is transmitted from the surrounding medium to the material or vice versa
- 2) heat transfer may cause an increase in the **thermal internal energy**
- 3) heat transfer takes place if there is a difference in temperature
- 4) heat transfer takes place from the side of higher temperature to the side of lower temperature

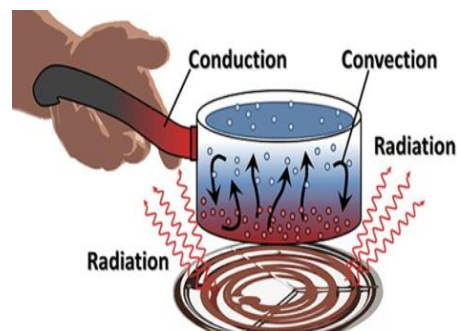
Mechanisms of Heat Transfer

We have talked about conductors and insulators, materials that permit or prevent heat transfer between bodies. Now let's look in more detail at rates of energy transfer.

In the kitchen you use a metal or glass pot for good heat transfer from the stove to whatever you're cooking, but your refrigerator is insulated with a material that prevents heat from flowing into the food inside the refrigerator. **How do we describe the difference between these two materials?**

The three mechanisms of heat transfer are **conduction, convection, and radiation**. **Conduction** occurs within a body or between two bodies in contact.

Convection depends on motion of mass from one region of space to another. **Radiation** is heat transfer by electromagnetic radiation, such as sunshine, with no need for matter to be present in the space between bodies.

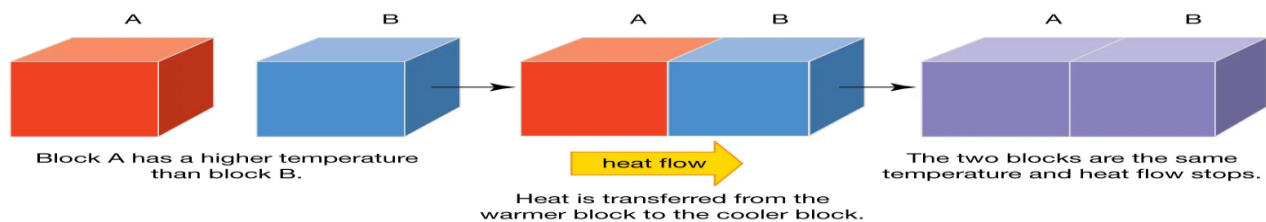


1- Conduction

If you hold one end of a copper rod and place the other end in a flame, the end you are holding gets hotter and hotter, even though it is not in direct contact with the flame. Heat reaches the cooler end by **conduction** through the material.

Remember that

Heat transfer occurs only between regions that are at different temperatures, and the direction of heat flow is always from higher to lower temperature



2- Convection

Convection is the transfer of heat by mass motion of a fluid (liquid or gas) from one region of space to another. Such as

- The cooling system of an automobile engine
- The flow of blood in the body
- Convection in the atmosphere plays a dominant role in determining the daily weather, and convection in the oceans is an important global heat-transfer mechanism
- Hot air rises and cool air falls to take its place (Convection occurs when warmer areas of a liquid or gas rise to cooler areas in the liquid or gas)
- Occurrence of storms and tornados

3- Radiation

Radiation is the transfer of heat by electromagnetic waves such as visible ight, infrared, and ultraviolet radiation.

All objects radiate energy continuously in the form of electromagnetic waves produced by thermal vibrations of the molecules.